

120 Days of Quantitative Finance

Week 7: Time Series Analysis Basics

Day 47: Volatility Estimation

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1. Introduction

Volatility is one of the most fundamental concepts in quantitative finance. It measures the variability of asset returns and serves as the primary proxy for risk.

In financial markets:

- Higher volatility implies greater uncertainty.
- Volatility directly impacts option pricing.
- Risk management frameworks rely heavily on volatility estimates.

Mathematically, volatility is defined as the standard deviation of returns.

2. Returns Definition

Let S_t denote the asset price at time t .

2.1 Simple Returns

$$r_t = \frac{S_t - S_{t-1}}{S_{t-1}}$$

2.2 Log Returns

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

Log returns are preferred in quantitative modeling because:

- They are time-additive.
- They align with continuous-time modeling (GBM).

3. Historical Volatility Estimation

Given a return series $r_1, r_2, \dots, r_n$:

3.1 Sample Mean

$$\bar{r} = \frac{1}{n} \sum_{t=1}^n r_t$$

3.2 Sample Variance

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2$$

3.3 Sample Volatility

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2}$$

This is referred to as **historical volatility**.

4. Annualized Volatility

If returns are measured daily, volatility must be scaled to annual units.

Assuming independent returns:

$$Var \left(\sum_{t=1}^T r_t \right) = T\sigma^2$$

Thus:

$$\sigma_{annual} = \sigma_{daily} \times \sqrt{252}$$

where 252 represents the average number of trading days in a year.

5. Rolling (Time-Varying) Volatility

Markets exhibit time-varying risk. To capture this, we compute rolling volatility.

Let m denote the rolling window length.

$$\sigma_t = \sqrt{\frac{1}{m-1} \sum_{i=0}^{m-1} (r_{t-i} - \bar{r})^2}$$

This provides a dynamic estimate of conditional risk.

6. Realized Volatility

For high-frequency data, realized volatility over interval T is:

$$RV_T = \sum_{i=1}^n r_i^2$$

where r_i are intraday returns.

Realized volatility is widely used in empirical finance and volatility forecasting.

7. Volatility Clustering

A stylized fact of financial markets:

$$Corr(r_t^2, r_{t-k}^2) > 0$$

This implies:

- Large shocks tend to be followed by large shocks.
- Small shocks tend to be followed by small shocks.

Returns themselves may show:

$$Corr(r_t, r_{t-k}) \approx 0$$

But squared returns exhibit persistence.

8. Conditional Heteroskedasticity

Volatility is not constant over time:

$$\text{Var}(r_t|\mathcal{F}_{t-1}) = \sigma_t^2$$

where $\mathcal{F}_{t-1}$ represents past information.

This motivates ARCH and GARCH models:

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2$$

Volatility becomes dependent on past shocks.

9. Financial Interpretation

Volatility plays a critical role in:

- Black-Scholes option pricing
- Value-at-Risk (VaR)
- Expected Shortfall
- Portfolio optimization
- Leverage sizing
- Risk parity strategies

Understanding volatility dynamics is essential for both pricing and risk management.

10. Conclusion

Volatility estimation forms the foundation of modern quantitative finance. While simple historical volatility assumes constant variance, empirical evidence shows that volatility is time-varying and clustered.

This motivates more advanced models such as:

- ARCH / GARCH
- Stochastic volatility models
- Implied volatility frameworks

Accurate volatility modeling is central to risk-aware financial decision making.